MIL-C-48021 (MU)
6 August 1973
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

CARTRIDGE, 107MM, HE: M453

HARDWARE FOR

1. SCOPE

1.1 This specification covers the hardware for one type of cartridge designated as Cartridge, 107MM, HE: M453.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Lot Numbering of Ammunition

FSC: 1315

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

A-9229760 - Hardware (For use with Cartridge, 107MM, HE: M453)

B-8781634 - Gage, Plug, Thread
A-8827845 - Gage, Indicator Groove
A-8841247 - Gage, Plug, Plain Cyl.
A-8841248 - Data Plate
D-8841249 - Gage, Concentricity
C-8841250 - Gage, Location
C-9202018 - Gage, Length
C-9202019 - Gage, Snap, Fixed
C-9202020 - Gage, Flush Pin
B-9202021 - Gage, Snap, Adj, Blade Type
D-9202025 - Gage, Flush Pin
D-9202027 - Gage, Location
B-9202028 - Gage, Flat Plug
A-9202098 - Data Plate
B-9202100 - Gage, Template
C-9202224 - Gage, Fixed Snap
D-9202225 - Gage, Snap
D-9202227 - Gage, Location
D-9202231 - Gage, Concentricity
C-9202232 - Gage, Concentricity
B-9202233 - Gage, Fixed Snap
B-9202234 - Gage, Length
B-9202235 - Gage, Template

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-64 - Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Hardware.-The hardware shall comply with all requirements specified on applicable drawings, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.4 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.4)

3.5 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-STD-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.
f. Number of units of product inspected.
g. Results obtained by code number, for all inspections performed.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1.) The First Article sample shall consist of the following:

a. Twenty-five (25) forward plates.
b. Twenty-five (25) liners.
c. Twenty-five (25) base plugs (uncoated).
d. Twenty-five (25) caps.
e. Twenty-five (25) covers.
f. Five (5) base plugs (coated).

All parts and completed assemblies submitted shall have been produced by the Contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, the following tests shall be performed:

4.2.2.1 Salt Spray Test of Base Plug.-Five (5) coated base plugs shall be subjected to the test specified in 4.4.1.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article

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quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any part or assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plan for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Plate, forward (See Dwg. C-9211935).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	(See Para 4.3.3.2)		
101.	Largest outside diameter	Gage	01001
102.	Total length	C9202224	01002
103.	Distance from largest diameter to step face	D9202225	01003
104.	Distance between bottom of step and opposite face (groove face)	Gage	01004
105.	Diameter of O-ring groove	Gage	01005
106.	Diameter of grooved face (second largest diameter)	Gage	01006
Major			
107.	Flat missing from circumference (between chamfer and ring groove)	Visual	01007
Minor:	(See Para 4.3.3.2)		
201.	Symmetry of face groove with largest outside diameter Gage	01008	
202.	Depth of face groove	B9202028	01009
203.	Parallelism of face groove with step face	D9202227	01010
Minor			
204.	Part damaged (edge chipped or nicked)	Visual	01011
205.	Evidence of poor workmanship (See 3.3)	Visual	01012

4.3.2.2 Liner (See Dwg. D-9211939).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Inside radius (as per dwg note)	Gage	02001

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	102. Liner damaged (punctured, cut, torn or dented)	Visual	02002
	103. Chamfer missing	Visual	02003
Minor	(See Para 4.3.3.3)		
	201. Total length	C9202018	02004
Minor			
	202. Evidence of poor workmanship (See 3.3)	Visual	02005

4.3.2.3 Plug, base (prior to coating) (See Dwg. D-9211941).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
	101. Pitch diameter of thread	Gage	03001
	102. Major diameter of thread	Gage	03002
	103. Inside radius of step, maximum	Gage	03003
	104. Diameter of flange	Gage	03004
	105. Diameter of ring groove	Gage	03005
	106. Diameter of press-fit surface	Gage	03006
	107. Diameter of body seat (largest diameter)	Gage	03007
	108. Width of driving boss	B9202233	03008
	109. Diameter of thread relief, minimum (min.)	B9202021	03009
	110. Length of driving boss from thread face	C9202020	02010
	111. Length to body seat from step face	B9202235	02011
	112. Width of ring groove	Gage	03012
	113. Distance between flange face and body seat	B9202234	03013
	114. Total length, maximum (max)	C9202019	03014
	115. Concentricity of largest outside diameter with press-fit diameter	C9202232	03015
	116. Concentricity of pitch diameter of thread with press-fit diameter including perpendicularity with body seat	D9202231	03016

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117.	Symmetry of driving boss with pitch diameter of thread	D9202027	03017
118.	Distance between body seat and circular boss adjacent to flange	D9202025	03018
119.	Distance between bottom of driving boss and bottom of step	Gage	03019
Minor			
201.	Radius of fillet from press-fit surface to body seat	Gage	03020
202.	Finish improper	Visual	03021
203.	Evidence of poor workmanship (See 3.3)	Visual	03022

4.3.2.4 Plug, Base (after coating) (See Dwg. D-9211941).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Thread damaged	Visual	04001
102.	Part damaged (crack, chipped, bent or distorted)	Visual	04002
Minor			
201.	Protective coating with bare spot	Visual	04003
202.	Evidence of poor workmanship (See 3.3)	Visual	04004

4.3.2.5 Cap, (See Dwg. C-8864935).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Pitch diameter of thread	B8781634	05001
102.	Minor diameter of thread, max	A8841247	05002
103.	Concentricity of thread with inside diameter	D8841249	05003

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104.	Diameter of thread relief (when applicable)	A8827845 A8841248	05004
105.	Location of thread relief (when applicable)	C8841250	05005
106.	Width of thread relief, max (when applicable)	Gage	05006
107.	Distance from base to datum diameter	Gage	05007
108.	Crack or perforation	Visual	05008
Major:	(See Para 4.3.3.1)		
109.	Inside diameter	Gage	05009
110.	Depth of bottom of cavity	B9202100	05010
111.	Large outside diameter	Gage	05011
112.	Wall thickness	A8202098	05012
113.	Base thickness	Gage	05013
Minor			
201.	Evidence of poor workmanship (See 3.3)	Visual	05014

4.3.2.6 Cover (See Dwg. C-9211945).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Crack or perforation	Visual	06001
102.	Burr	Visual	06002
Major:	(See Para 4.3.3.1)		
103.	Outside diameter	Gage	06003
104.	Thickness	Gage	06004
105.	Length	Gage	06005
Minor			
201.	Evidence of poor workmanship (See 3.3)	Visual	06006

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4.3.3 Testing.

4.3.3.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defectie part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The Contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.2 Dimensional control of cast parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5,000 parts. Individual cavity identification must be provided.

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If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shal be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

4.3.3.3 Dimensional control of plastic parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a stabilization time for the parts has been approved (see Note 1), a sample of five (5) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the stabilization time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity at the beginning of production and at intervals of 5,000 parts, or at the beginning and end of each shift, providing the number of parts produced during that shift is less than 5,000 parts.

The samples inspected for the beginning of production and the intervals described above shall be the first five (5) parts produced for the former and the last five (5) parts produced for the latter. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the

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cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

(NOTE 1) - In establishing a stabilization time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g.) every 30, 60, etc. mins.) until dimensional stability is attained. The stabilization time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the stabilization time shall be sent to Picatinny Arsenal, SARPA-QA-A-S. If any new change in material, process parameters, a reworked cavity, or new cavity is used, a new stabilization time shall be established and approved.

4.3.3.4 Salt spray test of base plug (See Dwg. D-9211941) Major Defect, Code No. 07001.

4.3.3.4.1 Batch type process.-When the finish is applied in a batch-type process, ten (10) of each part shall be selected from each batch and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the batch represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.4.2 Continuous-type process.-When the finish is applied in a continuous-type process, five (5) of each part shall be selected from each four (4) hours of production, or portion thereof, and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the quantity represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.1.

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4.3.3.4.3 Lot plan.-Ten (10) base plugs shall be selected from each lot and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the lot represented by the failing sample shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in 4.3.3 (see 6.3).

4.4 Test methods and procedures.

4.4.1 Salt spray.-The base plug shall be subjected to the salt spray test specified on the D-9211941 drawing using the equipment, method and procedures specified in ASTM B 117-64. (Part 7). The salt spray exposure time and the percentage of sodium chloride solution shall be as indicated on the drawing. Base plugs so tested shall not be returned to any lot. (Destructive test)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The hardware shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level C.-The hardware, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking.-Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

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a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which

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provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.5 Supersession data.-This specification includes the requirements of, and supersedes, Purchase Specification AEI-389, dated 21 February 1968.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1315-A003

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